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```

for i in range(1,6,1):
    for j in range(1,6,1):
        print(i*j, end=" ")
    print()

```

Initialization

$i = 01$

condition

$i < 6$

logic/code

print()
loop

update

$i = i + 1$

$i = 1$

$1 < 6$

$i = i + 1 = 2$

Initialization

condition

logic/code

update

$j = 1$

$j < 6$

$i*j$
 $i=1$

$j = j + 1$

$j = 1$

$1 < 6(T)$

1

$j = 1 + 1 = 2$

$j = 2$

$2 < 6(T)$

1

$j = 2 + 1 = 3$

$j = 3$

$3 < 6(T)$

1

$j = 3 + 1 = 4$

$j = 4$

$4 < 6(T)$

1

$j = 4 + 1 = 5$

$j = 5$

$5 < 6(T)$

1

$j = 5 + 1 = 6$

$j = 6$

$6 < 6(F)$

2

✓

$i = 2$

$2 < 6$

print()
loop

$j = 2 + 1 = 3$

Initialization	condition	logic/code	update
$i = 1$	$j < 6$	$i = 2, i \times 2$ $i = 1$	$j = j + 1$
$i = 1$	$1 < 6 (T)$	$2 \times 2 = 0$	$j = 1 + 1 = 2$
$i = 2$	$2 < 6 (T)$	$2 \times 2 = 0$	$j = 2 + 1 = 3$
$i = 3$	$3 < 6 (T)$	$2 \times 2 = 0$	$j = 3 + 1 = 4$
$i = 4$	$4 < 6 (T)$	$2 \times 2 = 0$	$j = 4 + 1 = 5$
$i = 5$	$5 < 6 (T)$	$2 \times 2 = 0$	$j = 5 + 1 = 6$
$i = 6$	$6 < 6 (F)$	$2 \times 2 = 0$	X

$i = 3$

$3 < 6$

print()
100 p

$i = 3 + 1 = 4$

Initialization	condition	logic/code	update
$j = 1$	$j < 6 p$	$i = 3, i \times 2$	$i = j + 1$
$j = 1$	$1 < 6 (T)$	$3 \times 2 = 1$	$j = 1 + 1 = 2$
$j = 2$	$2 < 6 (T)$	$3 \times 2 = 1$	$j = 2 + 1 = 3$
$j = 3$	$3 < 6 (T)$	$3 \times 2 = 1$	$j = 3 + 1 = 4$
$j = 4$	$4 < 6 (T)$	$3 \times 2 = 1$	$j = 4 + 1 = 5$
$j = 5$	$5 < 6 (T)$	$3 \times 2 = 1$	$j = 5 + 1 = 6$
$j = 6$	$6 < 6 (F)$	$3 \times 2 = 1$	X

$$i = 4$$

$$4 < 6 (T)$$

loop
print()

$$i = 4 + 1 = 5$$

Initialization

condition

logic/code

update/operation

$$j = 1$$

$$j < 6$$

$$j \% 2 \neq 0 \quad i = 4$$

$$j' = j + 1$$

$$j = 1$$

$$1 < 6 (T)$$

$$1 \% 2 \neq 0$$

$$j' = 1 + 1 = 2$$

$$j = 2$$

$$2 < 6 (T)$$

$$2 \% 2 = 0$$

$$j' = 2 + 1 = 3$$

$$j = 3$$

$$3 < 6 (T)$$

$$3 \% 2 \neq 0$$

$$j' = 3 + 1 = 4$$

$$j = 4$$

$$4 < 6 (T)$$

$$4 \% 2 = 0$$

$$j' = 4 + 1 = 5$$

$$j = 5$$

$$5 < 6 (T)$$

$$5 \% 2 \neq 0$$

$$j' = 5 + 1 = 6$$

$$j = 6$$

$$6 < 6 (F)$$

X

✓

$$i = 5$$

$$5 < 6 (T)$$

loop
print()

$$i = 5 + 1 = 6$$

Initialization

condition

logic/code

update/operation

$$j = 1$$

$$j < 6$$

$$i = 5 \quad i \% 2$$

$$j' = j + 1$$

$$j = 1$$

$$1 < 6 (T)$$

$$5 \% 2 = 1$$

$$j' = 1 + 1 = 2$$

$$j = 2$$

$$2 < 6 (T)$$

$$5 \% 2 = 1$$

$$j' = 2 + 1 = 3$$

$$j = 3$$

$$3 < 6 (T)$$

$$5 \% 2 = 1$$

$$j' = 3 + 1 = 4$$

$$j = 4$$

$$4 < 6 (T)$$

$$5 \% 2 = 1$$

$$j' = 4 + 1 = 5$$

$$j = 5$$

$$5 < 6 (T)$$

$$5 \% 2 = 1$$

$$j' = 5 + 1 = 6$$

$$j = 6$$

$$6 < 6 (F)$$

X

X



1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

for i in range (1, 6, 1)

for j in range (1, i+1, 1):

print(j, end=' ')

print()

Initialization

i = 1

i = 1

Condition

i < 6

1 < 6

loop/body

loop()
print()

update

j = j + 1

j = 1 + 1 = 2

Initialization

j = 1

condition

j < i + 1

loop/body

~~j = j + 1~~ print(j)

update

j = j + 1

j = 1

1 < 2

1

j = 1 + 1 = 2

j = 2

2 < 6 (T)

x

x

i = 2

2 < 6

loop()
print()

i = 2 + 1 = 3

Initialization

j = 1

condition

j < i + 1

j = 2 + 1

loop/body

print(j)

update

j = j + 1

j = 1

1 < 3 (T)

1

j = 1 + 1 = 2

j = 2

2 < 3 (T)

2

j = 2 + 1 = 3

j = 3

3 < 3 (F)

x

x



i=3

3 < 6 (True)

1100P

j = j + 1
3 + 1 = 4

Initialization

condition/logic

logic

update

j = 1

j < i + 1
j < 3 + 1 = 4 = j < 4

print(j)

j = j + 1

j = 1

1 < 4 (T)

~~print~~ 1 j = 1 + 1 = 2

j = 2

2 < 4 (T)

2 j = 2 + 1 = 3

j = 3

3 < 4 (T)

3 j = 3 + 1 = 4

j = 4

4 < 4 (F)

X

✓

i = 4 j = 4

4 < 6 (True)

loop

j = j + 1
4 + 1 = 5

Initialization

condition/logic

logic

update

j = 1

j < i + 1 = 4 + 1 = 5

print(j)

j = j + 1

j = 1

1 < 5 (T)

1

j = 1 + 1 = 2

j = 2

2 < 5 (T)

2

2 + 1 = 3

j = 3

3 < 5 (T)

3

3 + 1 = 4

j = 4

4 < 5 (T)

4

4 + 1 = 5

j = 5

5 < 5 (F)

X

X

i = 5

5 < 6 (True)

loop

j = i + 1
= 5 + 1

Initialization

condition / loop

logic

update

j = 1

j < i + 1 = 5 + 1 = 6

j = 1

1 < 6

1

j = j + 1
1 + 1 = 2

j = 2

2 < 6

2

2 + 1 = 3

j = 3

3 < 6

3

3 + 1 = 4

j = 4

4 < 6

4

4 + 1 = 5

j = 5

5 < 6

5

5 + 1 = 6

j = 6

6 < 6

X

X

i = 6

6 < 6 (false)

~~loop~~

X

ex:

1 2 3 4 5

1 2 3 4

1 2 3

1 2

1

for i in range(5, 0, -1):

for j in range(1, i + 1, 1):

print(j, end=" ")

print()

Initialization

i = 1

i = 5

condition

i > 0

1 > 0

logic

loop

loop

update

i = i + 1

i = 1 + 1 = 2



Initialization

$j = 1$

condition
 $j > 8 \quad j > i+1$

logic/c
 $\text{print}(j)$

update
 $j = j + 1$

$j = 2$

$1 > 6$

1

$1 + 1 = 2$

$j = 3$

$2 > 6$

2

$2 + 1 = 3$

$j = 4$

$3 > 6$

3

$3 + 1 = 4$

$j = 5$

$4 > 6$

4

$4 + 1 = 5$

$j = 6$

$5 > 6$

5

$5 + 1 = 6$

$j = 6$

$6 > 6$

X

X

$i = 4$

$4 > 0$ (True)

1000

$i = i - 1$
 $= 4 - 1$
 $= 3$

Initialization

$j = 1$

condition
 $j > i+1 > 5$

logic

update

$j = j + 1$

$j = 1$

$1 > 5$

1

$1 + 1 = 2$

$j = 2$

$2 > 5$

2

$2 + 1 = 3$

$j = 3$

$3 > 5$

3

$3 + 1 = 4$

$j = 4$

$4 > 5$

4

$4 + 1 = 5$

$j = 5$

$5 > 5$ (False)

X

X

$i = 3$

$2 > 0$ (True)

100p

$i = i - 1$
 $= 3 - 1 = 2$

Initialization

$j = 1$

$j = 1$

$j = 2$

$j = 3$

$j = 4$

condition

$j > i + 1$ ~~$j > 4$~~

$1 > 4$ (T)

$2 > 4$ (T)

$3 > 4$ (T)

$4 > 4$ (F)

logic

1

2

3

x

condition

~~$j = j + 1 = 2$~~

$j = 2 + 1 = 3$

~~$3 + 1 = 4$~~

~~$4 + 1 = 5$~~

x

$i = 2$

$2 > 0$ (True)

100p

$i = i - 1$
 $= 2 - 1 = 1$

Initialization

$j = 1$

$j = 2$

$j = 3$

$j = 4$

condition

$j > i + 1$ $j > 3$

$1 > 3$ (T)

$2 > 3$ (T)

$3 > 3$ (F)

logic

1

2

x

Update

$\text{print}(j)$ $j = j + 1$

$j = 1 + 1 = 2$

$j = 2 + 1 = 3$

x

$i = 1$

Initialization

$2 > 0$ (True)

100p

$i = i - 1$
 $= 2 - 1 = 1$

condition

$i > 2$

logic

$\text{print}(i)$

Update

$i = i - 1$

